SWINBURNE VIETNAM

HO CHI MINH CAMPUS

COS30018 - COS30018-Intelligent Systems

Week 10: Task C.5

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1. Overview of what I implemented

After completing Task C.4 (v0.3), the code base was upgraded to version v0.4 to solve more advanced forecasting problems:

1. Multistep prediction: predict a sequence of future closing prices for k days ahead.
2. Multivariate prediction: use multiple features (Open, High, Low, Close, Adj Close, Volume) as input to predict the future Close price.
3. Multivariate + multistep prediction: combine both—use multivariate inputs to predict k future Close prices.

I implemented new functions and modifications to my data processing and model pipeline to support these scenarios and validated them by running three modes: multistep, multivariate, and multivariate_multistep.

2. Implementation Summary (Functions Required by Task C.5)

2.1 Multistep prediction

Goal: predict the next k closing prices instead of only the next day.

To achieve this, I modified the dataset sequence-building logic so that the label y becomes a vector of length k:

- Single-step: y[i] is one value (Close at time t+1).
- Multi-step: y[i] becomes [Close(t+1), Close(t+2), …, Close(t+k)].

Key change: the model output layer must match the label shape:

- Dense(1) for single-step
- Dense(k) for multistep.

```
DENSE_UNITS = K_STEPS if (TASK5_MODE in ["multistep", "multivariate_multistep"] and K_STEPS > 1) else 1

model = build_recurrent_model(
    layer_name=MODEL_TYPE,
    input_shape=input_shape,
    num_layers=NUM_LAYERS,
    units=UNITS,
    dropout=DROPOUT,
    dense_units=DENSE_UNITS,
    optimizer="adam",
    loss="mean_squared_error",
)
```

2.2 Simple multivariate prediction

Goal: use multiple features of the same company as input to predict Close in the future.

I extended the feature list to include:

- Open, High, Low, Close, Adj Close, Volume

This makes the input tensor:

$$X \in \mathbb{R}^{(samples, lookback, n_{features})}$$

In my case, n_features = 6.

The target remains Close (single-step prediction), but now the model learns from a richer representation of market behavior including intraday range and trading activity.

2.3 Multivariate + multistep

Goal: combine the above two:

- input: multiple features (6 features)
- output: a sequence of k future Close prices

This produces:

- X_train shape = (samples, lookback, 6)
- y_train shape = (samples, k)
- model output layer = Dense(k)

(d) Dense(1) output for regression

Stock prediction here is regression (predicting a continuous price), so the output layer is Dense(1).

3. Less Straightforward Code Lines

3.1 Building multistep labels safely (indexing logic)

A key challenge was correctly indexing the future target window:

- define the start of prediction: target_start = i + forecast_horizon - 1
- define the end of prediction: target_end = target_start + (k_steps - 1)

This prevents out-of-bound indexing and ensures each training sample has a valid k-day label vector.

3.2 Keeping the model output compatible with y shape

For multistep, the model output must be a vector. That required:

- setting dense_units = k_steps
- ensuring predictions return shape (samples, k_steps)

This is the main conceptual difference between single-step and multistep regression models.

```python
DENSE_UNITS = K_STEPS if (TASK5_MODE in ["multistep", "multivariate_multistep"] and K_STEPS > 1) else 1

model = build_recurrent_model(
    layer_name=MODEL_TYPE,
    input_shape=input_shape,
    num_layers=NUM_LAYERS,
    units=UNITS,
    dropout=DROPOUT,
    dense_units=DENSE_UNITS,
    optimizer="adam",
    loss="mean_squared_error",
)
```

3.3 Inverse transform for multistep outputs (scaler handling)

Because the target scaler is fit on one column (Close), but the prediction is (samples, k), I used a reshape strategy:

1. flatten predictions to (-1, 1)
2. inverse transform with the scaler
3. reshape back to (samples, k)

This prevents errors and ensures predictions are converted back into real price units.

3.4 Handling “Adj Close” column differences in yfinance

Some tickers / cached datasets can return OHLCV without "Adj Close". To satisfy the assignment’s required feature list while maintaining compatibility, I implemented a safe fallback:

- if "Adj Close" is missing, use "Close" as a proxy and log a warning.

This ensures the code runs consistently and the input feature dimension remains correct

```python
    # --- Task C.5: ensure "Adj Close" exists if requested ---
    if "Adj Close" in feature_columns and "Adj Close" not in df.columns:
        if "Close" in df.columns:
            df["Adj Close"] = df["Close"]
            print("Warning: 'Adj Close' not found in data; using 'Close' as proxy for 'Adj Close'.")
        else:
            raise ValueError("Adj Close requested but neither 'Adj Close' nor 'Close' exists in downloaded data.")
```

3.5 Plotting multistep predictions clearly

When k > 1, plotting a full matrix (samples, k) results in multiple predicted lines. To keep the chart readable, I plotted only t+1 (the first horizon), and labeled it as t+1 of k.

4. References Used

- TensorFlow/Keras RNN documentation (stacking RNNs and output shapes):
  https://www.tensorflow.org/api_docs/python/tf/keras/layers/LSTM
  https://www.tensorflow.org/api_docs/python/tf/keras/layers/GRU
  https://www.tensorflow.org/api_docs/python/tf/keras/layers/SimpleRNN
- scikit-learn MinMaxScaler (fit/transform and inverse_transform):
  https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html
- yfinance documentation/examples (feature availability varies by ticker):
  https://pypi.org/project/yfinance/

5. Experiment Results Summary (Multistep / Multivariate / Combined)

I tested three modes on CBA.AX with the same lookback window (60 days) and evaluated results using MAE/MSE/RMSE. A **saved terminal** with all the tested results have been save in **Result-ScreenShots folder.**

5.1 Mode A — Multistep (univariate, Close only, k=5)

- Output: Next 5-day prediction vector
- Reported metrics (from logs):
  - MAE: 2.1189
  - RMSE: 2.6793

```
Train time (sec): 14.456616200011922
8/8 ━━━━━━━━━━━━━━━━━━━━ 1s 41ms/step
Test MAE: 2.11897080131199
Test MSE: 7.178915917984027
Test RMSE: 2.6793499058510495
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
Next 5-day predictions for CBA.AX: [115.758705 116.03598  117.67698  115.335075 117.03559 ]
```

5.2 Mode B — Multivariate (single-step, 6 features)

- Output: Next-day prediction scalar
- Reported metrics:
  - MAE: 2.8345
  - RMSE: 3.4370

```
Train time (sec): 15.16313369999989
8/8 ━━━━━━━━━━━━━━━━━━━━ 1s 44ms/step
Test MAE: 2.8344562364661154
Test MSE: 11.813147154279582
Test RMSE: 3.4370259170218054
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
Next-day prediction for CBA.AX: 113.85897827148438
```

5.3 Mode C — Multivariate + Multistep (6 features, k=5)

- Output: Next 5-day prediction vector
- Reported metrics:
  - MAE: 4.9349
  - RMSE: 5.6959

```
Train time (sec): 14.449672799994005
8/8 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step
Test MAE: 4.934886574123214
Test MSE: 32.44320758464921
Test RMSE: 5.695893923226556
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
Next 5-day predictions for CBA.AX: [110.30679  110.908104 111.13529  111.10187  109.715096]
```

Mode	Input Type	Output	k	MAE	RMSE
Multistep	Close only	Next k Close prices	5	2.1189	2.6793
Multivariate	OHLC + Adj Close + Volume	Next-day Close	1	2.8345	3.437
Multivariate + Multistep	OHLC + Adj Close + Volume	Next k Close prices	5	4.9349	5.6959

Discussion

- The combined problem (multivariate + multistep) is the hardest, so higher MAE/RMSE is expected because it must learn richer inputs while forecasting multiple steps ahead.
- Multistep forecasting also increases uncertainty because errors can grow across horizons.

6. Conclusion

Task C.5 successfully extended the project to v0.4 by supporting:

- multistep prediction (predict k future days),
- multivariate prediction (use multiple features as input),
- multivariate multistep prediction (combine both).

The implementation required careful handling of label construction, output shape alignment (Dense(k)), correct inverse-transform logic for multistep outputs, and practical data column compatibility (Adj Close). The experiments demonstrate that the system can run all three modes and produce valid evaluation metrics.